FULL STACK PROJECT REPORT

TITLES & DETAILS:

- Project Title: Rudraksha Computers – Institute Management System
- Student Name: Anushka Kolhe
- Roll No: 05
- Course Name: Full Stack
- College Name: K. K. Wagh Institute of Engineering Education and Research, Nashik
- Guide Name:
    - ❖ Prof. P. D. Rakibe
- Academic Year: 2025-2026

ABSTRACT:

The Rudraksha Computers – Institute Management System is a web-based application developed to manage and automate the operations of a computer training institute. It is built using HTML, CSS, JavaScript for the frontend, Node.js with Express for the backend, and MongoDB as the database.The system includes two main modules: the Owner Module, which manages students, courses, attendance, and study materials, and the Student Module, which allows students to access courses, download notes, track progress, and view attendance. Students can mark topics as completed, and progress is updated dynamically. A certificate is generated upon 100% course completion.

This system reduces manual work, improves efficiency, and provides a user-friendly platform for both administrators and students.

INTRODUCTION:

The Rudraksha Computers – Institute Management System is a web-based application designed to automate and simplify the management of a computer training institute. Traditional manual methods are time-consuming and error-prone, so this system provides a digital solution for efficient handling of data.

The system is developed using HTML, CSS, JavaScript for the frontend, Node.js with Express for the backend, and MongoDB as the database. It includes two main modules: the Owner Module for managing students, courses, attendance, and notes, and the Student Module for accessing courses, tracking progress, downloading notes, and generating certificates.

The main aim of this project is to improve efficiency, reduce manual work, and provide a centralized platform for managing institute activities.

Client Details:

- Client Name: Gayatri More
- Organization Name: Rudraksha Computers
- Address: Dagunana Nagar More Near Niphad Phata Pimpalgaon B
- Location (Link): https://goo.gl/maps/FrGcoQzmjjxvoDnf6?g_st=aw
- Contact: +91 8484834055
- Email: gayatrinikhilmore@gmail.com
- Type of Business/Service: Teaching Computers Various Courses

PROBLEM STATEMENT:

At Rudraksha Computers, the management of students, courses, attendance, and study materials is currently handled using manual or basic methods, which are time-consuming and prone to errors. There is no centralized system to efficiently manage all activities in one place. Students face difficulties in accessing learning materials, tracking their progress, and staying updated with course content. Similarly, the administrator finds it challenging to maintain records, monitor performance, and generate reports or certificates.
Therefore, there is a need for a web-based institute management system that can automate these processes, improve efficiency, and provide a better experience for both students and the institute management.

REQUIREMENT ANALYSIS:

Functional Requirements:

- **User Authentication:** Separate login system for Admin and Students using JWT.
- **Student Management:** Owner can add, update, delete, and manage student records efficiently.
- **Course Management**: Owner can create and manage courses with details like duration and content.
- **Content Management:** Course content is divided into topics; each topic can include notes and completion tracking.
- **Notes Upload System:** Owner can upload study materials (PDF, DOCX, PPTX) for each course topic.
- **Continue Learning Feature:** Students can resume their course and view structured content for each topic.
- **Download Notes:** Students can download notes uploaded by the owner for their enrolled courses
- **Certificate Generation:** Students can download certificates after completing 100% of the course.

Non-Functional Requirements

- **Performance:** The system should load quickly and respond efficiently to user actions without delays.
- **Usability:** The interface should be simple, user-friendly, and easy to navigate for both owner and students.
- **Secure Authentication (JWT):** The application uses JSON Web Token (JWT) for secure login and role-based access for admin and students**.**
- **Fast Performance:** React ensures fast rendering and smooth user experience through its virtual DOM.
- **Scalable Architecture:** The system is designed in a way that new features and users can be added easily in the future.
- **Efficient Data Handling:** MongoDB allows flexible and efficient storage and retrieval of data such as users, enquiries, and homework.

TECHNOLOGY STACK:

Frontend

- **Technologies :** HTML, CSS, JavaScript
- **Language:** JavaScript
- **Styling:** Custom CSS (Tailwind CSS, Flexbox/Grid)

Backend (Backend as a Service)

- **Runtime:** Node.js
- **Framework:** Express.js
- **Authentication:** JSON Web Token (JWT)
- **Middleware:** CORS, dotenv

Database (Supabase - PostgreSQL)

- **Database:** MongoDB
- **ODM:** Mongoose

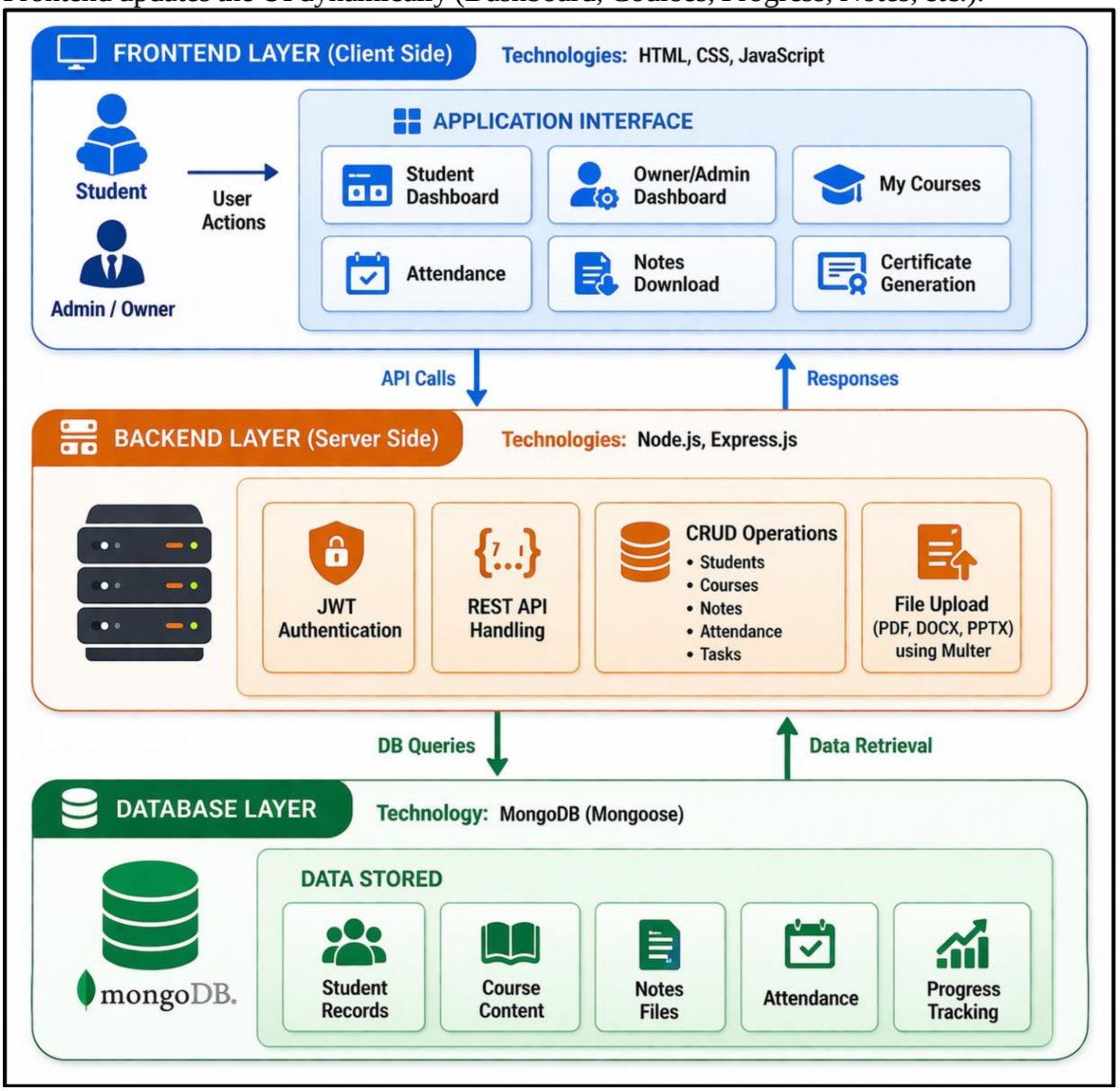
SYSTEM ARCHITECTURE:

The system follows a **client-server architecture**.

Flow:

- User interacts with the **Frontend (HTML, CSS, and JavaScript)**.
- Frontend sends **HTTP requests (API calls)** to the backend using Fetch API.
- Backend (**Node.js + Express**) processes requests”
    - Authenticates the user using **JWT (JSON Web Token)**.
    - Performs **CRUD operations** (Students, Courses, Content, Notes, and Attendance).
    - Handles **file uploads** (PDF, DOCX, PPTX) using Multer.
    - Interacts with **MongoDB database** using Mongoose.

- Frontend updates the UI dynamically (Dashboard, Courses, Progress, Notes, etc.).



IMPLEMENTATION DETAILS:

Frontend Implementation

- **Pages/Components Developed:**
  1. **Home Page:** Displays institute information, available courses, and navigation option
  2. **Login & Signup Pages:** Provides secure authentication interface for Owner and Students.
  3. **Owner Dashboard:** Manage students, courses, packages, attendance, notes, and daily tasks.
  4. **Student Dashboard:** View enrolled courses, progress, attendance, and important update
  5. **My Courses Page:** Displays enrolled courses with progress tracking and continue learning option.

6. **Continue Learning Page:** Shows course content in structured format with download notes and mark complete options

- **UI Design:** Implemented using **HTML, CSS, and JavaScript** with responsive layout, clean structure, and user-friendly navigation. Flexbox and Grid are used for proper alignment and design consistency.

Backend Implementation

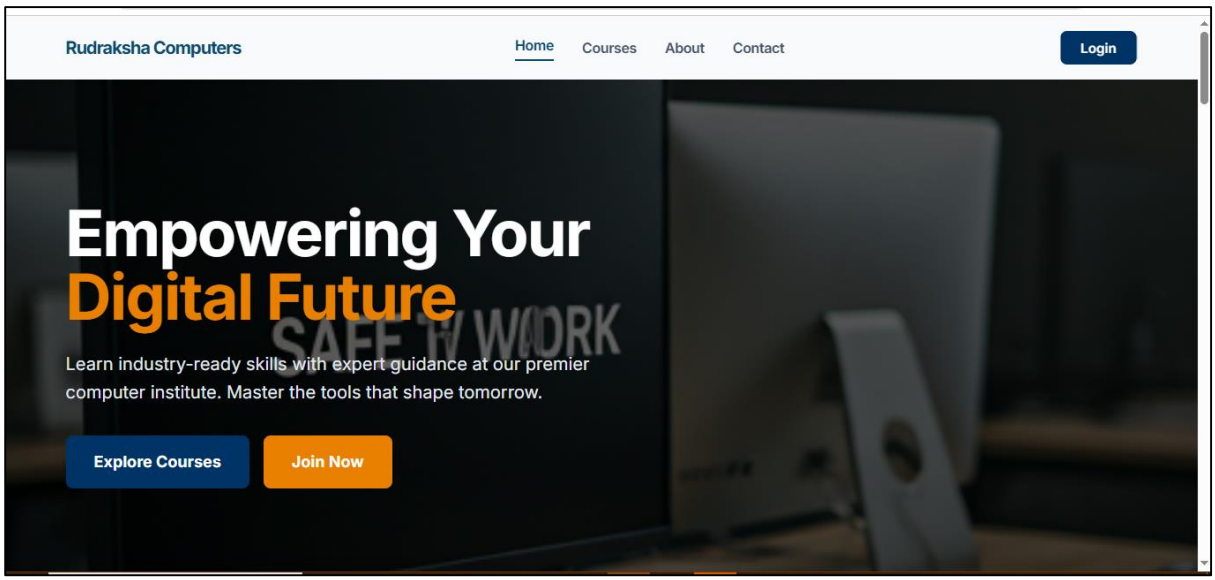
- **Developed REST APIs using Express.js** to handle all client-server communication
- Implemented **JWT-based authentication** for secure login system for Owner and Students.
- **Created routes for:**
  - ❖ **Authentication:** Login and registration for Owner and Students
  - ❖ **Students:** Add, update, delete, and manage student details
  - ❖ **Courses:** Create, update, delete courses and manage course content
  - ❖ **Notes:** Upload and manage notes (PDF, DOCX, PPTX)
  - ❖ **Attendance:** Mark and track student attendance
  - ❖ **Tasks/Daily Tasks:** Add and manage daily learning tasks
  - ❖ **Progress:** Track course completion and update progress
  - ❖ **Certificates:** Generate certificate after course completion

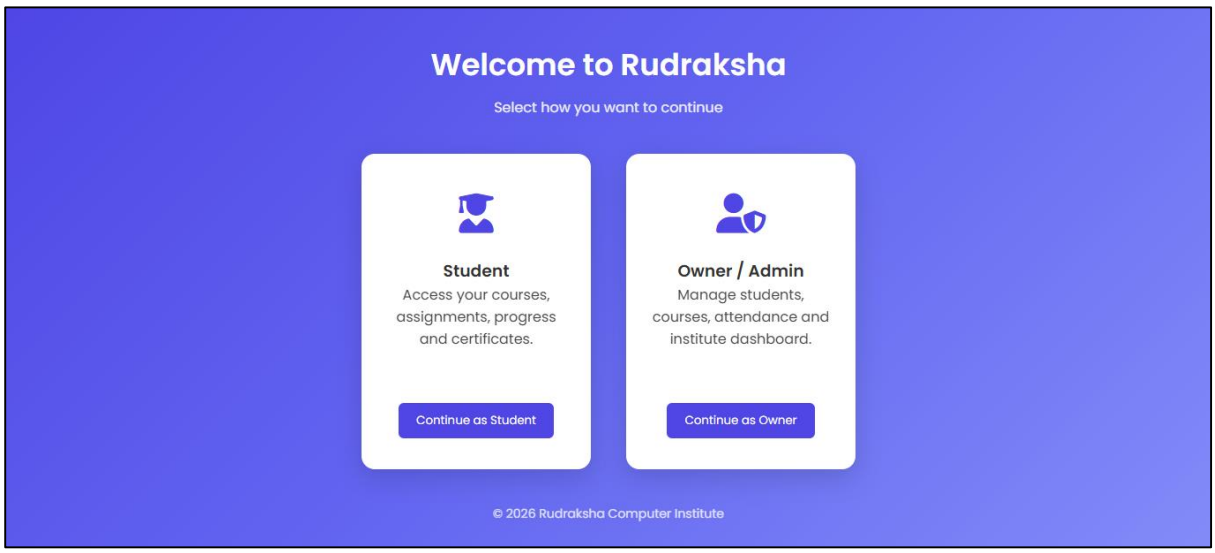
  Middleware used :
  - ❖ Authentication middleware (JWT verification)
  - ❖ Error handling middleware
  - ❖ CORS and JSON parsing middleware
  - ❖ Database Implementation
- **MongoDB used for storing all application data**
- **Collections:**
  - **users:** Stores Owner (Admin) and Student details
  - **students:** Student information, enrolled courses, progress, and fees
  - **courses:** Course details, description, content, and notes
  - **attendance:** Records of student attendance
  - **tasks:** Daily tasks assigned to students
  - **notes:** Uploaded files (PDF, DOCX, PPTX) linked to courses
  - **certificates:** Generated certificates after course completion

- **Mongoose used for schema design and data validation**

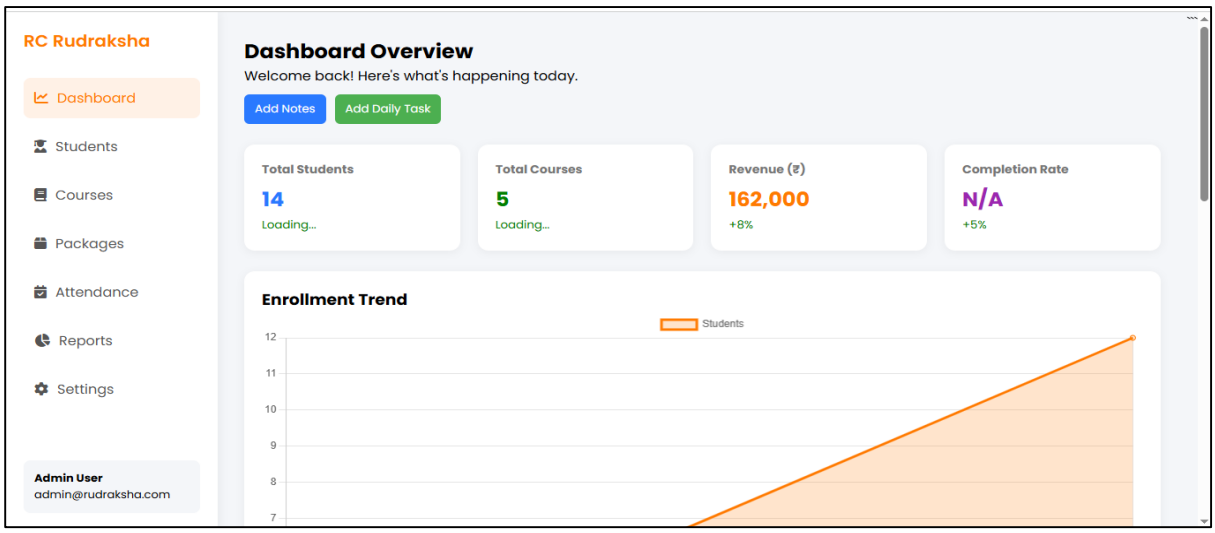
SCREENSHOTS:

Homepage + Login Modal

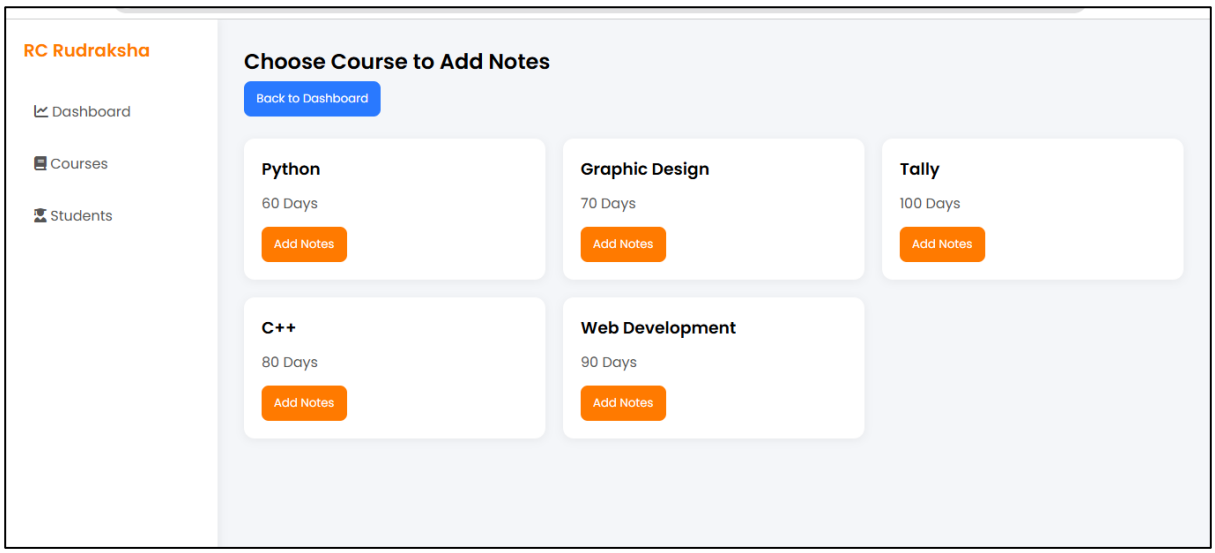


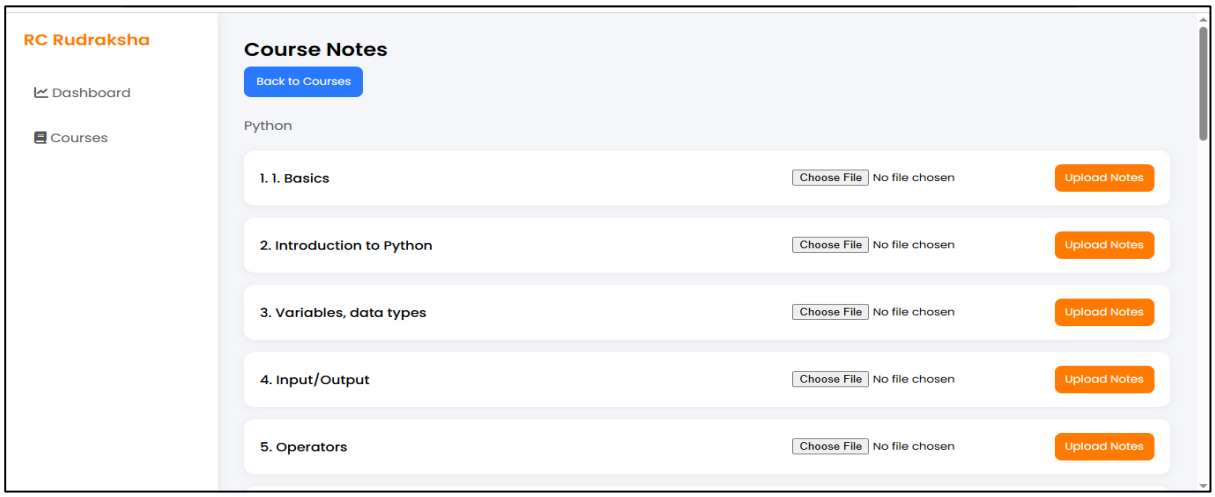


Owner Module: All activities on Owner Dashboard

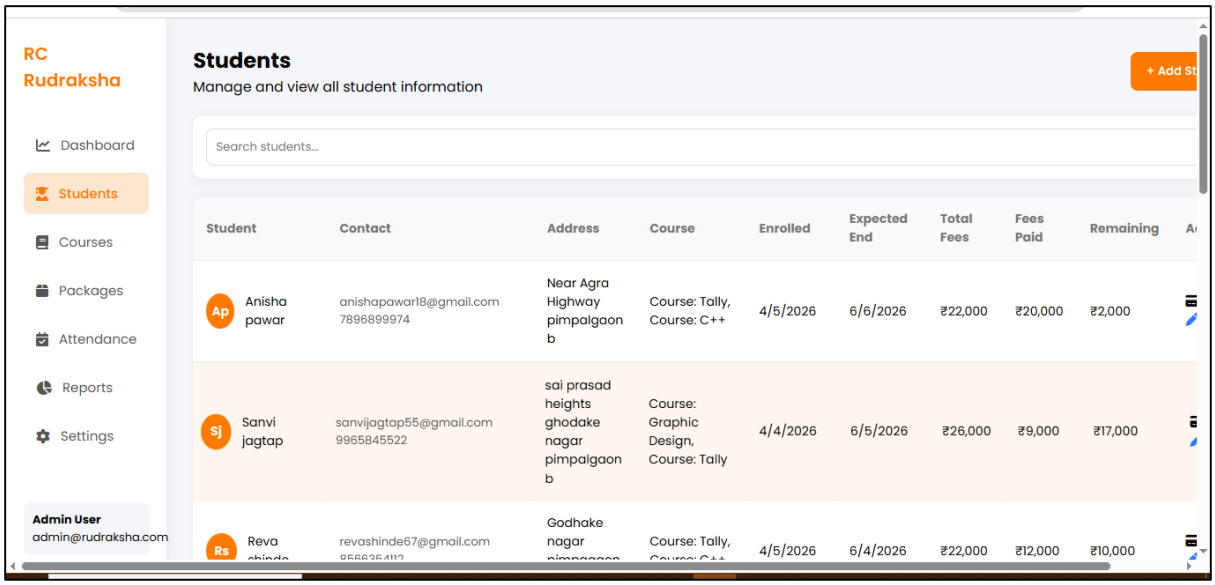


Add Notes: Here Owner can add notes

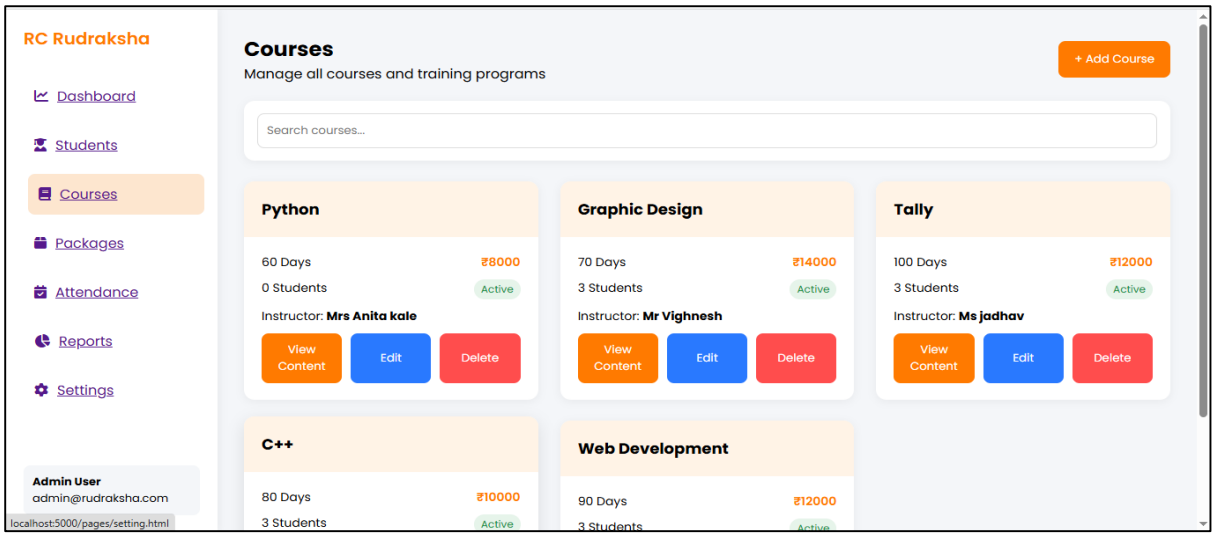




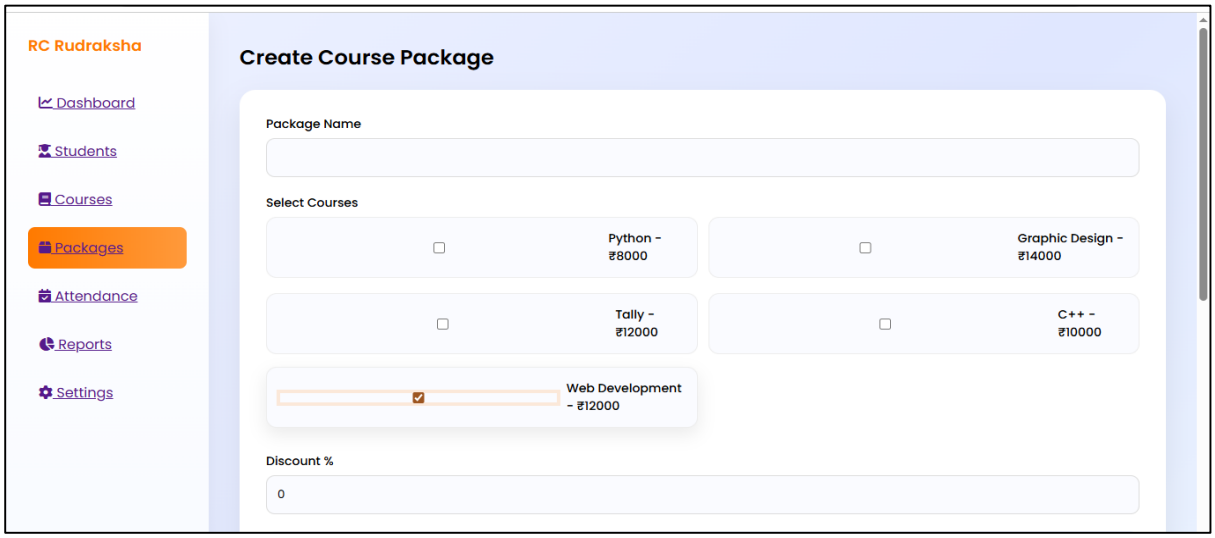
Students: Owner Can tracks and Manage students.



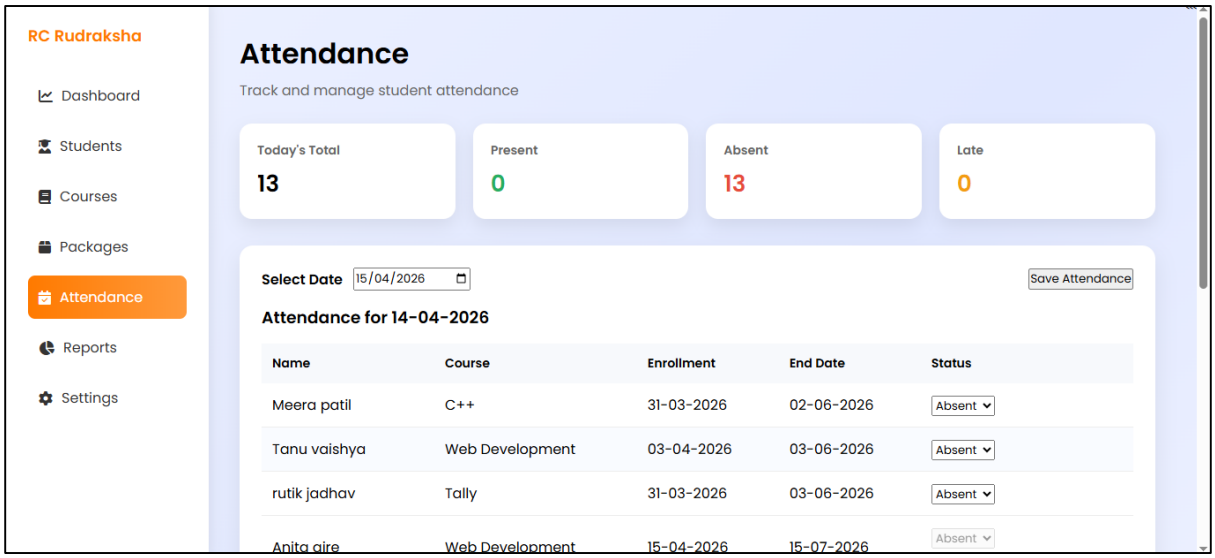
Courses: Here owner can add,Edit,Delete and View content of coureses.



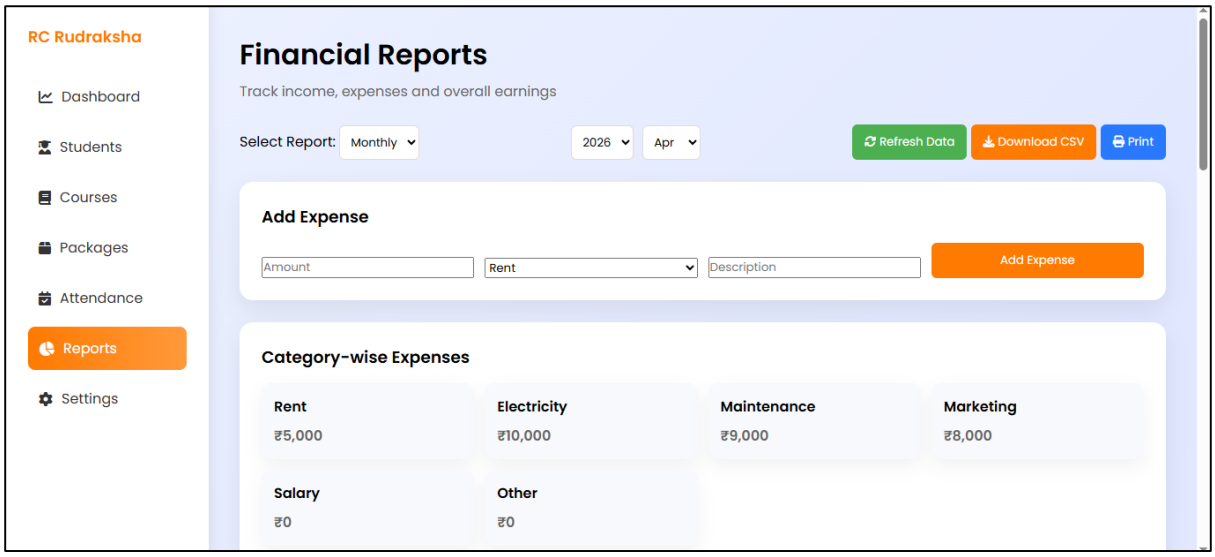
Packages: Owner can create Packages with Discount



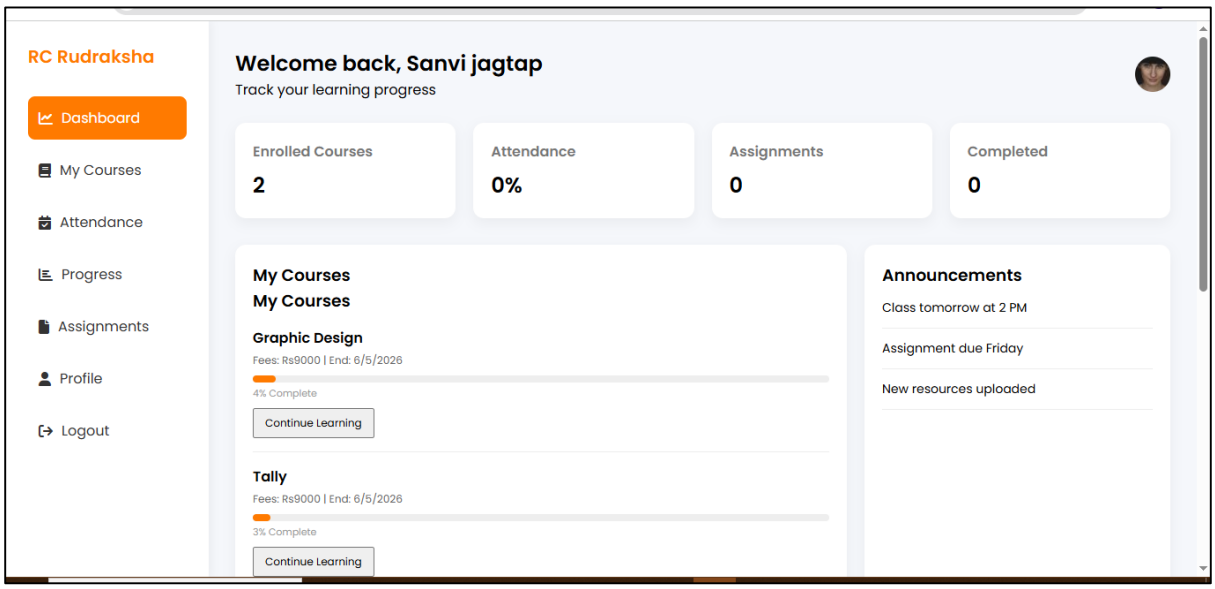
Attendance: Owner can Track attendance



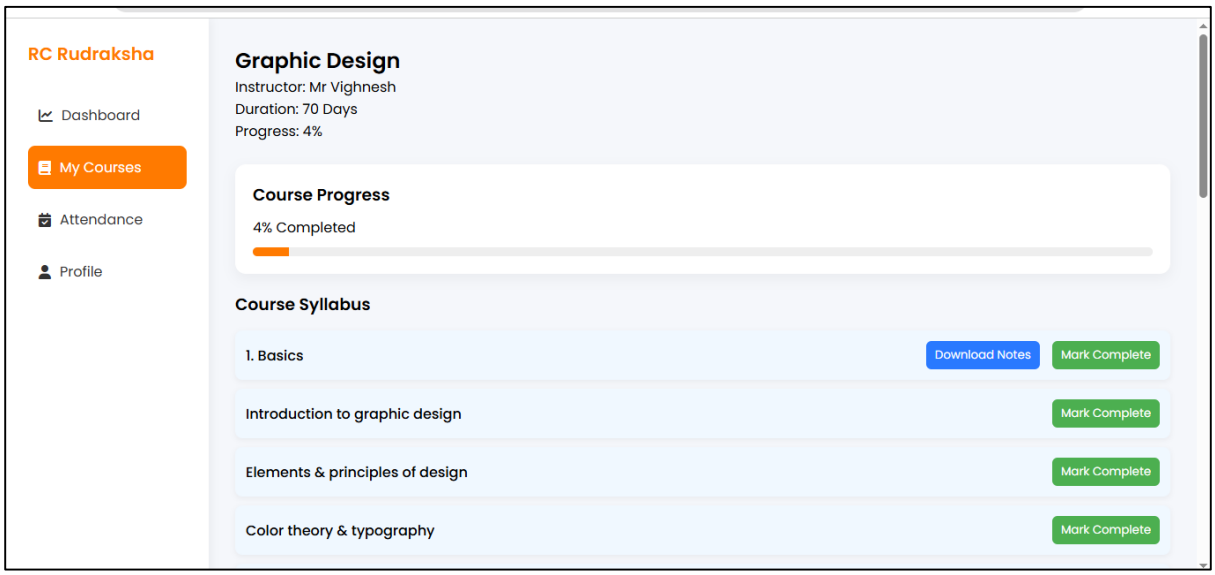
Financial Reports : Owner Can track his expenses and incomes from here



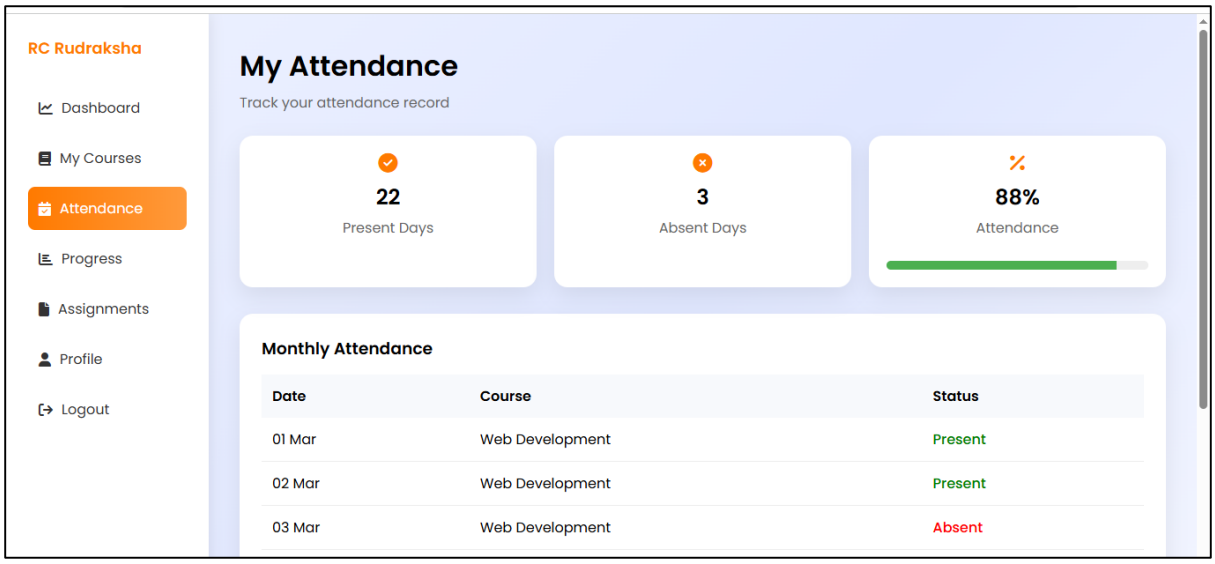
Student Module: Student Activities and progress



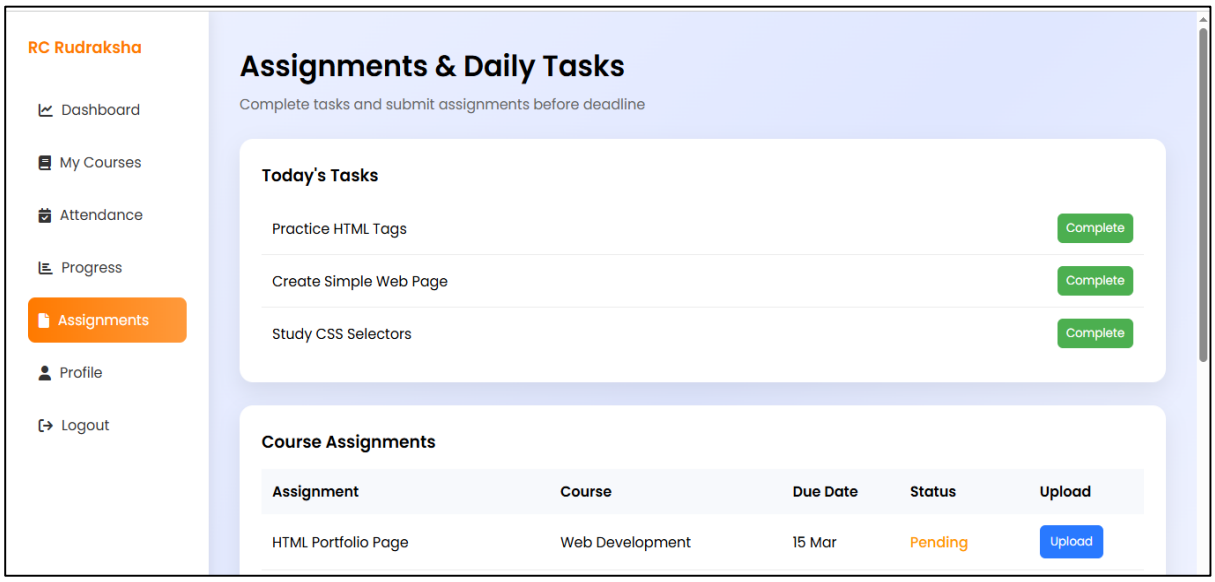
My courses: Student can see his enrolled courses from here



Attendance: Student can see his or her attendance from here



Assignments: Student can track daily work and assignment from here



API TESTING:

- **Tool Used:** Postman & Browser Network Tab
- **CRUD Operations Verified:**
  - ❖ **CREATE:** Added new students, courses, notes (PDF/DOC/PPT), and daily tasks from Owner Dashboard. Verified data is stored correctly in MongoDB.
  - ❖ **READ:** Loaded Student Dashboard and Owner Dashboard. Verified data such as courses, notes, attendance, and student details are fetched correctly from the database
  - ❖ **UPDATE:** Updated student details, course content, and progress status (mark complete). Verified changes are reflected instantly in both database and UI.
  - ❖ **DELETE:** Deleted students, courses, or notes from Owner Dashboard. Verified data is removed properly from the database.

- **File Upload Test:** Uploaded notes (PDF, DOCX, PPTX) from Owner Module. Verified that: Files are stored in server uploads folder. Files are accessible and downloadable in Student Module

DEPLOYMENT DETAIL S:

Frontend URL (Live):

Backend URL (API):

Database: MongoDB Atlas (Cloud-hosted)

FINAL OUTPUT/RESULT:

The **Rudraksha Computers – Institute Management System** has been successfully designed, developed, and implemented. The system is fully functional and meets the requirements of the client.

- Authentication system using JWT is implemented for secure login of Owner and Students
- Owner Dashboard is fully functional (manage students, courses, packages, notes, attendance, tasks)
- Student Dashboard is fully functional (view courses, continue learning, download notes, track progress, attendance, assignments)
- Course content management with notes upload and download feature is working properly
- Progress tracking system updates correctly across dashboard and course pages
- Certificate generation feature is implemented after course completion
- File upload system (PDF, DOCX, PPTX) for notes is implemented successfully
- Application provides a clean, user-friendly, and responsive interface

Final Cost Analysis (Rudraksha Computers)

Cost Category	Description	Estimated Cost (₹)
Development Cost	Based on COCOMO II (5 PM × ₹12,000)	60,000
Domain Name	Website domain (1 year)	800
Hosting (Frontend – Vercel)	Free / basic plan	0
Backend Hosting (Render)	Basic server deployment	3,000

Database (MongoDB Atlas)	Free tier / minimal usage	0
Internet & Electricity	Development + testing usage	2,000
Software & Tools	VS Code, Postman (Free)	0
Maintenance Cost (1 Year)	Minor updates, bug fixes	7,000
Miscellaneous Costs	Testing, setup, small expenses	2,000
Total		**74,800**

CLIENT FEEDBACK:

"The Rudraksha Computers – Institute Management System has significantly improved the way we manage our institute operations. Managing students, courses, attendance, and study materials is now much easier and faster. The dashboards are simple, clean, and very user-friendly. Features like course-wise notes upload, student progress tracking, and certificate generation are especially useful. Overall, the system saves time and reduces manual work."

– **Gayatri More, Owner, Rudraksha Computers**

GEOTAGGED PHOTO WITH CLIENT:

Discussion and Final Deployment Demonstration of Rudraksha Computers System with Client.

LIMITATIONS:

- **Limited Analytics and Reporting:** The system provides basic features like course progress and student management, but lacks advanced analytics such as performance dashboards, progress graphs, and detailed reports, which can enhance decision-making.
- **No Real-Time Notifications:** The system does not support real-time updates (like live alerts). Adding technologies like WebSockets can improve instant communication between owner and students

FUTURE SCOPE:

- Enhance attendance system with detailed reports and automated tracking features
- Implement parent dashboard for monitoring student progress, attendance, and performance
- Integrate email/SMS/WhatsApp notifications for updates like attendance, tasks, and announcements
- Add online test/quiz system for evaluating student performance
- Develop mobile application for easier access on smartphones

CONCLUSION:

This full-stack project helped in understanding real-world application development by integrating frontend, backend, and database technologies. I successfully gathered client requirements, designed the system, developed a responsive frontend using HTML, CSS, and JavaScript, built backend APIs using Node.js and Express, and integrated MongoDB for data storage.The project provided hands-on experience in authentication using JWT, API integration, CRUD operations, file handling, and dashboard development. Features like course management, notes upload/download, progress tracking, attendance, and certificate generation were implemented successfully. The final system is user-friendly, efficient, and reduces manual work by providing a complete digital solution for managing the institute effectively.

REFERENCES:

- **Frontend Documentation:** https://rw3schools/
- **Node.js Documentation:** https://nodejs.org/
- **Express.js Docs:** https://expressjs.com/
- **MongoDB Docs:** https://www.mongodb.com/docs/
- **Tailwind CSS Docs:** https://tailwindcss.com/docs
- **GitHub Repository:**